



AI-Based Network Intrusion Detection and Threat Intelligence Dashboard

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Abstract

AI-Based Network Intrusion Detection and Threat Intelligence Dashboard detects and classifies suspicious network activities using Machine Learning and Deep Learning models such as Random Forest and DNN. It helps security teams monitor network traffic, identify attacks, and view threat information through an interactive dashboard.

- Main benefit: Enables faster threat detection, monitoring, and response to network security attacks.

Problem Statement & Objectives

- Problem: Traditional network security systems may fail to detect evolving and complex cyber attacks efficiently, making continuous monitoring and threat analysis difficult.
- Objective 1: Detect malicious and suspicious network traffic using AI/ML models.
- Objective 2: Classify detected network activities into different attack types.
- Objective 3: Provide a dashboard to monitor threats, alerts, and security insights.
- Scope: Network traffic analysis, intrusion detection, attack classification, threat monitoring, and dashboard visualization

Existing vs Proposed System

a	Existing System	Proposed System
Threat Detection	Rule/signature based	AI/ML based
Detection	Mainly known attacks	Known and suspicious patterns
Analysis	Manual/limited	Automated
Alerts	Limited	Automated threat alerts
[Limitation / Gap]	Limited detection & manual analysis	Automated AI-based detection & analysis

Feasibility Study

- Technical:Uses Python, AI/ML models, and web technologies that are easy to develop in VS Code.
- Economic:Uses open-source tools and libraries, so development cost is low.
- Operational:Provides a simple dashboard for easy threat monitoring and analysis
- Schedule:Can be completed within the semester with proper planning and testing.

Results & Testing

Results: Successfully detects and classifies network traffic using AI/ML models and displays detected threats on the dashboard.

Testing: Tested data preprocessing, model predictions, intrusion detection, dashboard functions, and database connectivity.

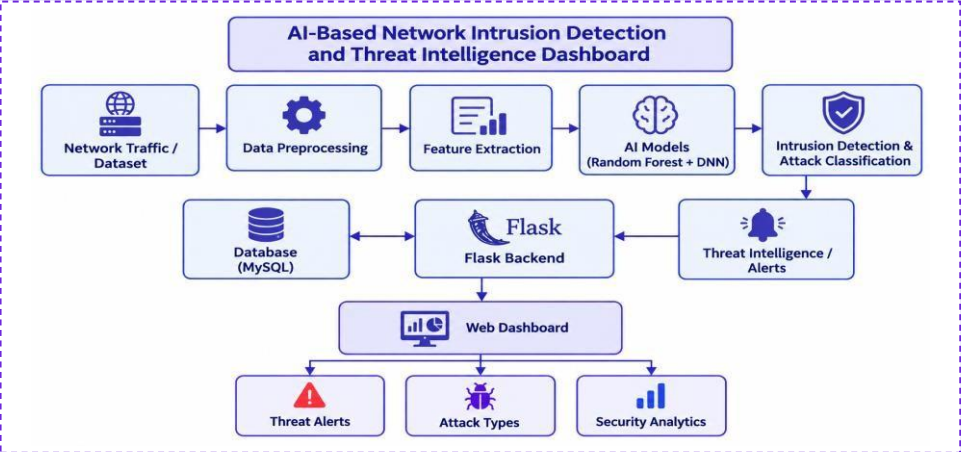
Conclusion

- The proposed system successfully uses AI/ML techniques to detect and classify network intrusions.
- It helps improve automated and efficient network security monitoring.

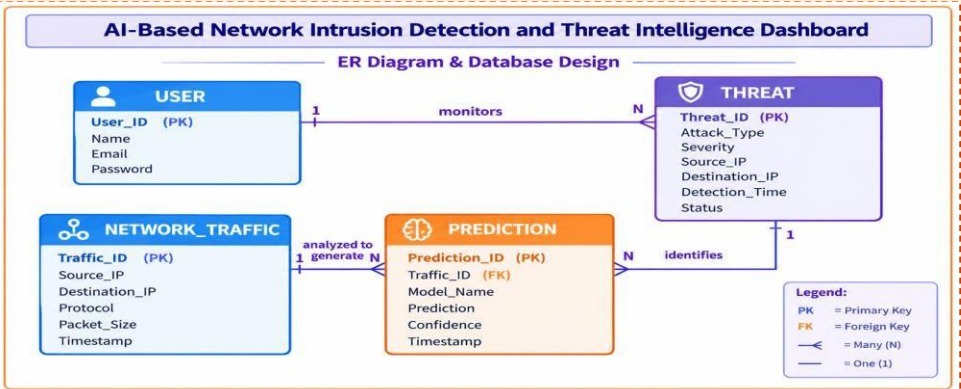
Tools & Technologies

- Front-End: HTML, CSS, JavaScript
- Back-End:Python
- Database: MySQL / MongoDB
- Tools: Git & GitHub,, VS Code

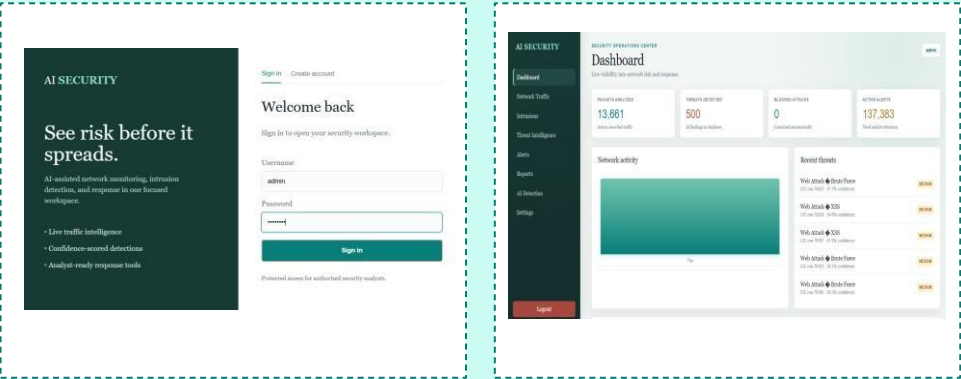
System Architecture



ER Diagram & Database Design



Implementation Screenshots



Future Scope

Real-Time Detection: Integrate live network traffic monitoring and advanced AI models for faster detection of new and complex attacks.

Advanced Security: Add automated alerts and external threat-intelligence feeds for improved threat analysis and response.